UniGAD: Unifying Multi-level Graph Anomaly Detection

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Motivation

- Existing GAD methods focus on a single object type
  - Node-level, edge-level, or graph-level detection but never all together
- Real-world anomalies span multiple levels
  - A money-laundering transaction involves both an abnormal account AND its broader community
- No unified framework exists
  - Current approaches miss cross-level correlations critical for detection

Key Challenges

- Challenge 1: How to unify multi-level formats?
  - Node / edge / graph tasks differ inherently
  - LLM methods need text attributes - unavailable in GAD
  - k-hop graph prompts erase anomaly information

- Challenge 2: How to unify multi-level training?
  - Gradient directions conflict across task levels
  - Naive joint training degrades all levels
  - Existing multi-task GNNs lack multi-label supervision

Framework Overview

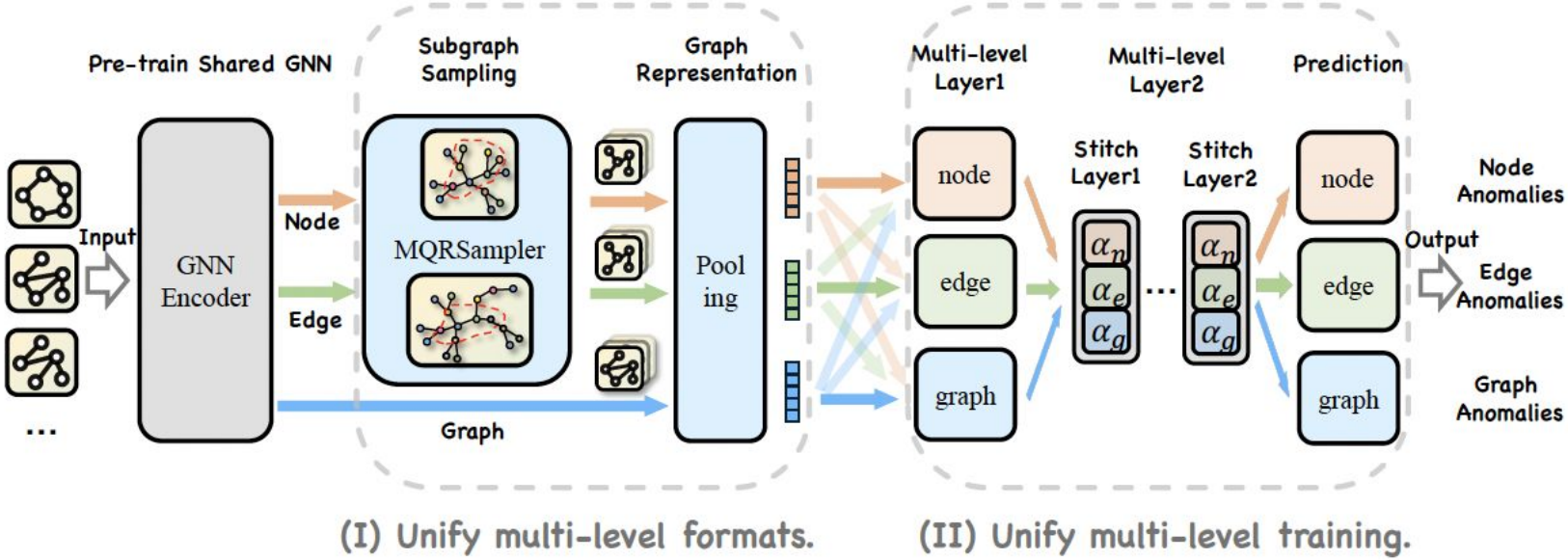


Figure 1: The overall framework of UniGAD.

MRQSampler: Spectral Subgraph Sampling

- Higher Rayleigh Quotient (RQ) = more anomalous. MRQSampler finds the subgraph that maximizes RQ.

1. Rooted Subtree
   a. Sample from rooted subtrees (k-depth)
   b. Aligns with GNN message-passing
2. Maximize RQ
   a. Add node only if delta(v) exceeds the current RQ(S)
      i. delta (v): score of the candidate addition relative to S
      ii. RQ(S): score of the current whole subgraph
   b. Theorem 1 & Theorem 2
3. DP Algorithm
   a. Bottom-up dynamic programming
   b. O(N log N) complexity
   c. Fully parallelizable

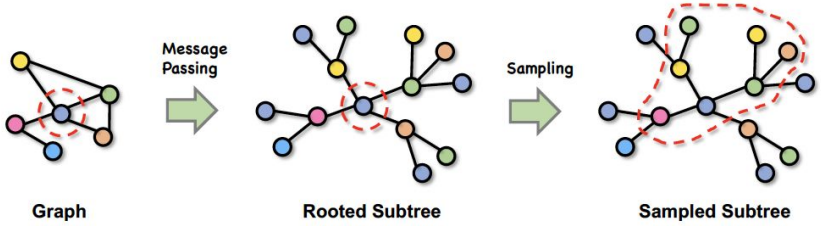


Figure 2: Message passing in GNNs and rooted subtree sampling.

MRQSampler: Spectral Subgraph Sampling

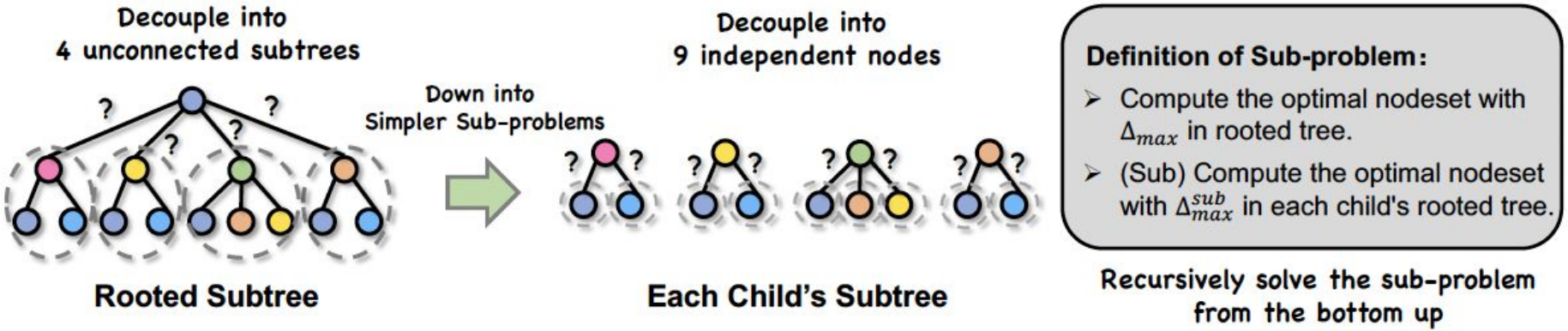


Figure 3: MRQSampler: (i) Derive the condition (Theorem 2) satisfied with the optimal subtree. (ii) Decompose the problem into simpler sub-problems by recursing through the tree depth to solve the optimal subtree with the dynamic programming (DP) algorithm.

GraphStitch Network: Multi-level Training

- Node level, Edge level, Graph level
  - Separate but Identical Network
- GraphStitch Unit - Learnable α Matrix (3×3)
  - Linear combination of embeddings across all levels

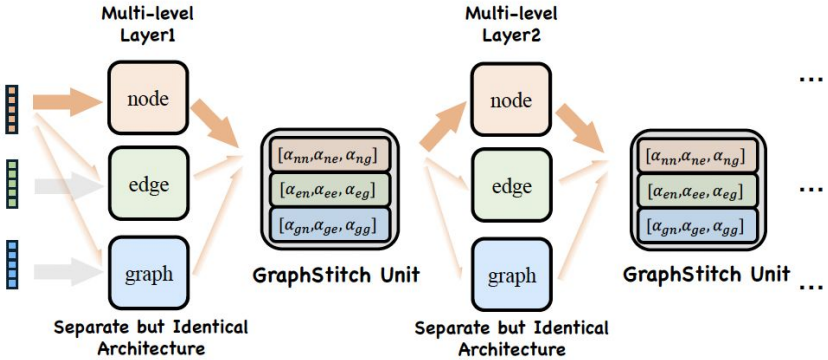


Figure 4: GraphStitch network structure in UniGAD. Node level is highlighted.

- Adaptive Sharing
  - α weights control cross-level borrowing
  - Cross-terms = 0 -> independent training
- Gradient Surgery
  - Projects gradients onto normal plane
  - Prevents interfering gradient components
- Fault Tolerance
  - Missing labels -> that level's influence zeroed
  - Still receives info from other levels

$$(\tilde{\mathbf{e}}_N, \tilde{\mathbf{e}}_E, \tilde{\mathbf{e}}_G) = diag\left[\begin{pmatrix} \alpha_{nn} & \alpha_{ne} & \alpha_{ng} \\ \alpha_{en} & \alpha_{ee} & \alpha_{eg} \\ \alpha_{gn} & \alpha_{ge} & \alpha_{gg} \end{pmatrix} (\mathbf{e}_N, \mathbf{e}_E, \mathbf{e}_G)\right].$$

Framework Overview

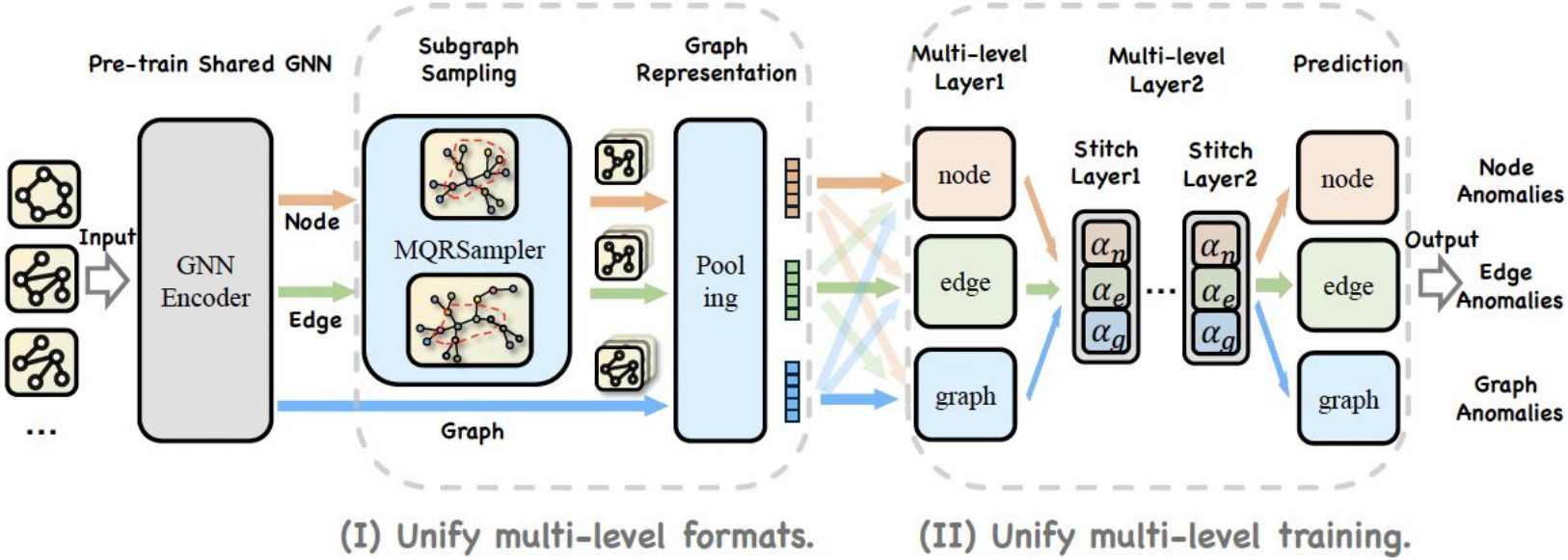


Figure 1: The overall framework of UniGAD.

Experimental Setup

- Datasets (14 total)
  - Single-Graph (7): Reddit, Weibo, Amazon, Yelp…
  - Multi-Graph (7): BM-MN/MS/MT, MUTAG, MNIST…
  - Node + edge labels on single-graph datasets
  - Node + graph labels on multi-graph datasets
  - New: T-Group (37K graphs, 11M nodes)
    - If a node appears in 3 malicious social groups -> anomalous
- Baselines (17 methods)
  - Node-level (9): GCN, GIN, GraphSAGE, GAT, BWGNN…
  - Edge-level (9): concatenated variants (GCNE, GINE…)
  - Graph-level (6): OCGIN, GLocalKD, RQGNN…
  - Multi-task (2): GraphPrompt-U, All-in-One-U
    - Since other methods were not originally proposed for joint multi-task training
- Metrics
  - AUROC, Macro F1, AUPRC

Table 1: Detailed statistics of the datasets used in our experiments.

Dataset	Train%	# Graphs	# Edges	# Nodes	# Dims	Nodes_{ab}	Edges_{ab}	Graphs_{ab}
Reddit	40%	1	168,016	10,984	64	3.33%	2.72%	/
Weibo	40%	1	416,368	8,405	400	10.33%	5.71%	/
Amazon	70%	1	8,847,096	11,944	25	6.87%	2.49%	/
Yelp	70%	1	7,739,912	45,954	32	14.53%	13.89%	/
Tolokers	50%	1	530,758	11,758	10	21.82%	33.44%	/
Questions	50%	1	202,461	48,921	301	2.98%	7.50%	/
T-Finance	40%	1	21,222,543	39,357	10	4.58%	2.77%	/
BM-MN	40%	700	40,032	12,911	1	48.91%	/	14.29%
BM-MS	40%	700	30,238	9,829	1	31.99%	/	14.29%
BM-MT	40%	700	32,042	10,147	1	34.49%	/	14.29%
MUTAG	40%	2,951	179,732	88,926	14	4.81%	/	34.40%
MNIST0	10%	70,000	41,334,380	4,939,668	5	35.46%	/	9.86%
MNIST1	10%	70,000	41,334,380	4,939,668	5	35.46%	/	11.25%
T-Group	40%	37,402	93,367,082	11,015,616	10	0.64%	/	4.26%

Multi-level Performance: Node + Edge (AUROC)

RQ1: How effective is UniGAD in unified multi-level anomaly detection? (node-level and edge-level)

- UniGAD outperforms all single-level methods in nearly every dataset
- Unified training boosts each level via cross-level information sharing

Table 2: Comparison of unified performance (AUROC) at both node and edge levels with different single-level methods, multi-task methods, and our proposed method.

	Dataset	Reddit		Weibo		Amazon		Yelp		Tolokers		Questions		T-Finance	
	Task-level	Node	Edge	Node	Edge	Node	Edge	Node	Edge	Node	Edge	Node	Edge	Node	Edge
Node-Level	GCN	62.60	/	97.97	/	82.37	/	57.62	/	75.21	/	70.15	/	90.70	/
	GIN	65.59	/	95.64	/	92.17	/	74.46	/	75.15	/	69.13	/	86.43	/
	GraphSAGE	62.25	/	94.45	/	84.53	/	82.12	/	79.74	/	72.47	/	78.16	/
	SGC	52.12	/	97.71	/	80.24	/	53.03	/	69.51	/	70.59	/	74.21	/
	GAT	65.87	/	94.40	/	96.24	/	77.40	/	78.90	/	71.38	/	90.60	/
	BernNet	66.68	/	93.93	/	96.62	/	81.48	/	76.68	/	70.28	/	92.37	/
	PNA	65.28	/	97.43	/	81.41	/	71.81	/	75.82	/	71.78	/	68.17	/
	AMNet	68.31	/	94.17	/	97.31	/	81.42	/	76.67	/	68.63	/	93.58	/
	BWGNN	64.65	/	97.42	/	97.80	/	83.11	/	80.51	/	70.25	/	96.03	/
Edge-level	GCNE	/	63.10	/	99.03	/	78.63	/	57.80	/	73.59	/	79.05	/	87.63
	GINE	/	67.36	/	98.09	/	79.74	/	67.58	/	69.27	/	80.75	/	79.05
	GSAGEE	/	**67.52**	/	98.67	/	78.92	/	73.30	/	**76.98**	/	**87.51**	/	77.14
	SGCE	/	53.36	/	98.55	/	76.41	/	52.02	/	70.59	/	74.24	/	69.01
	GATE	/	67.07	/	97.92	/	90.20	/	72.96	/	71.92	/	81.64	/	83.09
	BernE	/	65.57	/	97.87	/	89.60	/	73.93	/	73.39	/	84.78	/	87.80
	PNAE	/	64.15	/	99.10	/	75.71	/	67.98	/	75.09	/	84.05	/	83.91
	AME	/	66.73	/	97.08	/	89.36	/	73.69	/	71.99	/	84.93	/	86.19
	BWE	/	67.39	/	98.93	/	91.61	/	75.63	/	75.66	/	85.00	/	92.27
Multi-task	GraphPrompt-U	50.03	49.78	55.29	50.71	50.01	50.96	49.83	49.56	51.24	49.66	55.16	50.01	OOT	OOT
	All-in-One-U	51.35	54.10	48.61	52.63	56.11	54.80	49.77	49.13	50.41	49.29	51.49	64.24	OOT	OOT
UniGAD (Ours)	UniGAD - GCN	**71.65**	65.46	99.02	**99.13**	82.92	80.04	63.22	61.74	77.26	72.89	**73.92**	74.72	95.68	93.75
	UniGAD - BWG	64.42	53.60	**99.07**	99.10	**97.84**	**92.18**	**86.23**	**79.05**	**80.62**	74.85	70.97	73.45	**96.49**	**94.32**

Multi-level Performance: Node + Graph (AUROC)

RQ1 continued. (node-level and graph-level)

- Strong node performance boost graph performance
- Shared encoder avoids redundant computation
- Memory usage comparable to node-level models

Table 3: Comparison of unified performance (AUROC) at both node and graph levels with different single-level methods, multi-task methods, and our proposed method.

	Dataset	BM-MN		BM-MS		BM-MT		MUTAG		MNIST0		MNIST1		T-Group	
	Task-level	Node	Graph	Node	Graph	Node	Graph	Node	Graph	Node	Graph	Node	Graph	Node	Graph
Node-level	GCN	86.31	/	90.17	/	92.30	/	99.38	/	94.10	/	93.84	/	91.81	/
	GIN	56.73	/	50.41	/	54.90	/	99.39	/	93.55	/	93.49	/	61.51	/
	GraphSAGE	50.00	/	50.00	/	49.95	/	99.26	/	**99.99**	/	**99.99**	/	64.15	/
	SGC	50.27	/	50.87	/	49.44	/	89.19	/	86.97	/	86.97	/	82.55	/
	GAT	58.47	/	62.52	/	65.72	/	99.42	/	99.90	/	**99.99**	/	78.17	/
	BernNet	60.06	/	65.58	/	59.18	/	98.97	/	**99.99**	/	**99.99**	/	93.85	/
	PNA	72.96	/	55.19	/	75.61	/	98.76	/	99.80	/	99.87	/	55.66	/
	BWGNN	93.05	/	87.22	/	88.97	/	99.50	/	**99.99**	/	**99.99**	/	94.81	/
Graph-level	OCGIN	/	98.46	/	81.97	/	58.05	/	89.50	/	57.24	/	86.15	/	64.53
	OCGTL	/	98.48	/	83.17	/	59.99	/	92.19	/	59.35	/	93.45	/	46.77
	GLocalKD	/	92.36	/	77.25	/	53.23	/	72.77	/	66.69	/	57.42	/	78.53
	iGAD	/	91.68	/	96.68	/	99.14	/	96.28	/	98.93	/	99.50	/	64.44
	GmapAD	/	50.00	/	50.00	/	50.00	/	75.48	/	OOM	/	OOM	/	OOM
	RQGNN	/	**98.79**	/	97.98	/	99.83	/	96.41	/	96.62	/	95.57	/	73.90
Multi-task	GraphPrompt-U	51.59	46.85	50.54	48.67	51.42	49.38	97.08	68.23	81.16	83.88	81.37	6.16	47.40	50.81
	All-in-One-U	67.87	3.21	54.70	19.42	69.70	45.89	50.63	48.98	OOT	OOT	OOT	OOT	OOT	OOT
UniGAD (Ours)	UniGAD - GCN	**99.75**	94.29	**99.60**	**99.67**	**99.63**	**99.99**	99.50	96.33	97.93	98.99	98.11	99.59	95.57	88.73
	UniGAD - BWG	92.60	68.74	93.30	68.55	90.76	56.01	**99.54**	**96.73**	**99.99**	**99.61**	**99.99**	**99.98**	**96.19**	**88.78**

Zero-Shot Task Transferability

RQ2: Can UniGAD transfer across levels with no labels at test level during training?

- UniGAD beats all multi-task prompt methods in zero-shot settings
- Zero-shot performance surpasses supervised baselines on Yelp & BM-MS
- Demonstrates genuine cross-level knowledge transfer

Table 4: Zero-shot transferability (AUROC) at node and edge levels.

Methods	Reddit		Weibo		Amazon		Yelp		Tolokers		Questions		T-Finance	
	N→E	E→N	N→E	E→N	N→E	E→N	N→E	E→N	N→E	E→N	N→E	E→N	N→E	E→N
GraphPrompt-U	54.06	47.43	57.03	42.85	49.76	50.26	49.97	49.94	48.56	51.08	54.26	51.97	OOT	OOT
All-in-One-U	49.23	49.93	52.22	54.30	52.61	42.35	49.48	44.50	48.34	50.22	49.83	51.97	OOT	OOT
UniGAD - GCN	**59.67**	**59.46**	**98.31**	**98.59**	76.20	82.38	58.28	60.92	71.45	73.35	69.54	**65.37**	91.63	90.17
UniGAD - BWG	53.32	57.63	94.71	96.87	**82.64**	**96.41**	**75.56**	**84.08**	**74.04**	**78.49**	**71.02**	62.72	**93.60**	**95.68**

Table 5: Zero-shot transferability (AUROC) at node and graph levels.

Methods	BM-MN		BM-MS		BM-MT		MUTAG		MNIST0		T-Group	
	N→G	G→N	N→G	G→N	N→G	G→N	N→G	G→N	N→G	G→N	N→G	G→N
GraphPrompt-U	50.60	51.57	51.97	46.95	46.62	48.06	59.62	64.26	83.98	**88.06**	58.28	58.35
All-in-One-U	**94.39**	65.69	52.63	40.88	44.86	34.27	61.63	36.13	OOT	OOT	OOT	OOT
UniGAD - GCN	72.82	**87.63**	**81.49**	**90.83**	**62.85**	**79.26**	**72.79**	**88.53**	**85.24**	70.57	**86.86**	**75.89**
UniGAD - BWG	64.61	57.56	65.33	51.34	55.78	53.41	66.92	87.03	74.23	63.70	86.81	64.81

Ablation Study

RQ3: What are the contributions of each module in UniGAD?

- MRQSampler
  - No subgraph sampling
  - Simple sampler with 2 hop neighbours
  - Random sampling
  - Wrong sampling can be worse than none
- GraphStitch
  - Replacing with MLP drops multi-level scores
  - Stitch unit essential for balanced learning
- Both are necessary
  - Full model peaks on all metrics

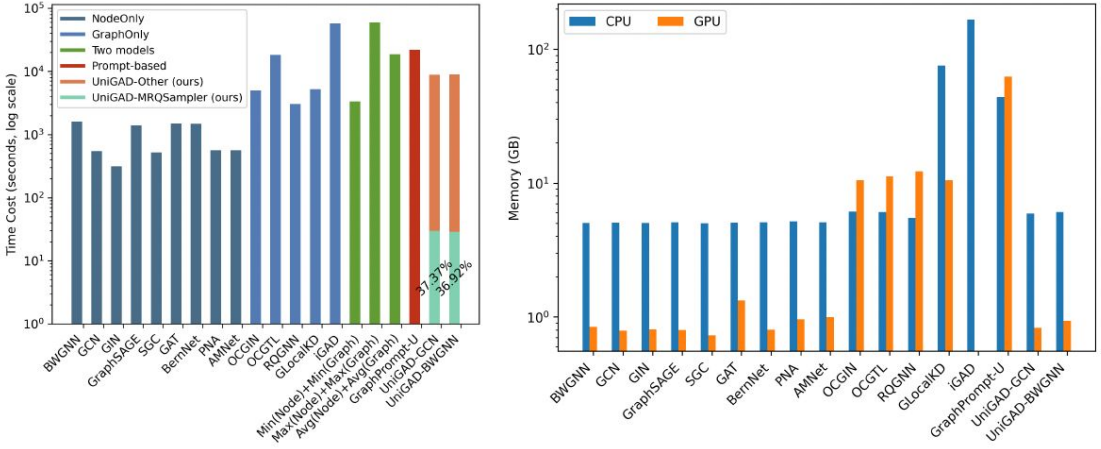
Table 6: Performance of UniGAD and its variants.

	BM-MS				Reddit			
Metrics	AUROC		Macro F1		AUROC		Macro F1	
Task-level	node	graph	node	graph	node	edge	node	edge
w/o GS.	97.13	98.99	80.35	95.79	68.69	66.06	53.83	52.78
w 2hop.	97.49	99.94	67.29	84.20	67.53	63.62	51.77	50.69
w RS.	93.85	84.92	85.88	72.21	65.32	61.85	52.32	51.03
w/o ST.	99.94	95.51	99.47	84.91	67.74	65.92	54.35	52.52
UniGAD	99.60	99.67	99.57	95.86	71.65	65.46	56.70	53.80

Efficiency Analysis

RQ4: How do time and space efficiencies compare? (T-Group dataset)

- Time: slower than combination of the single fastest models, faster than average of the combination
- Memory: on par with node-level models - far below graph-level & prompt methods
- Prompt methods time out on large datasets - UniGAD scales gracefully



(a) Comparing the execution time. (b) Comparing the peak memory usage.

Figure 5: The evaluation of time and space efficiency metrics. We highlight the percentage of total execution time spent by MRQSampler.

Key Contributions

- First Unified Multi-level GAD
  - Jointly detects node, edge, and graph anomalies
  - Addresses cross-level correlations previously ignored
- MRQSampler
  - Spectral sampling with theoretical optimality guarantee
- GraphStitch Network
  - Adaptive α-weighted cross-level sharing
- Strong Results + Zero-shot
  - SOTA across 14 datasets vs. 17 methods
  - Zero-shot surpasses supervised baselines in some cases

Limitations & Future Work

Current Limitations

- GNN encoder is mainly node-focused (GCN, BWGNN)
- May not fully capture edge and graph-level patterns
- Graph-level structure may be underrepresented

Future Directions

- Use more structure-aware encoders (beyond GCN/BWGNN)
- Better capture edge and graph-level patterns
- Incorporate more global graph structure

Conclusion

- UniGAD: First framework for joint node, edge & graph anomaly detection
  - MRQSampler
    - Spectral subgraph sampling with theoretical optimality guarantee
  - GraphStitch
    - Adaptive multi-level training via learnable α weights
- Results: SOTA on 14 datasets, 17 baselines, strong zero-shot transferability